
Blacktip Refine — AWS Marketplace Setup Guide

A complete, end-to-end walkthrough for deploying **Blacktip Refine** from the AWS Marketplace into **your own AWS account**. Refine is self-hosted: the software runs on one EC2 instance in your environment, your cost and inventory data never leave your account, and there is no SaaS to send data to.

***Time:** ~20-30 minutes, mostly an ~8-minute CloudFormation create. **Cost:** the Refine subscription is **\$0.00 on AWS Marketplace (Bring Your Own License)** — you pay only for the small EC2 instance (~\$30-60/mo on `t3.medium`) plus your separately-agreed Blacktip license. See pricing tiers in your order form.*

What gets deployed

The CloudFormation template provisions, in **your** account:

- **1 EC2 instance** (default `t3.medium`) running the Refine container, pulled from AWS Public ECR.
- **1 S3 bucket** that receives your signed license file.
- **1 IAM role** scoped to that bucket + AWS Systems Manager (for keyless admin access).
- **1 security group** (UI access locked to a CIDR you choose).
- Optional: an **Elastic IP** and an **HTTPS Application Load Balancer** (with your ACM certificate).

Refine connects **read-only** to your AWS accounts, Azure subscriptions, and GCP projects and ranks cost-optimization opportunities across all three clouds in one queue.

Prerequisites

- An AWS account with permission to launch CloudFormation (creates EC2, S3, IAM, SG).
- A VPC and a public subnet in your chosen region.
- The CIDR you'll allow to reach the Refine UI (your office/VPN egress IP, e.g. `203.0.113.10/32` — find yours at <https://checkip.amazonaws.com>).
- (Optional, for HTTPS) an ACM certificate in the same region + two subnets in different AZs for the load balancer.

Step 1 — Subscribe on AWS Marketplace

1. Open the Blacktip Refine listing and choose **View purchase options**.
2. Review the offer — **Contract total \$0.00** (Bring Your Own License). Accept the terms (the EULA is the **Download EULA(s)** link).
3. Choose **Subscribe**. You'll land on the **Purchase confirmation** page once the agreement is **Active**.

Step 2 — Launch the software

1. On the confirmation page, choose **Launch your software**.

2. On the Launch page, under **Service**, select **ECS** simply to proceed.

Important: Refine deploys as an EC2 host via the bundled CloudFormation template — **not** as an ECS/EKS task. Use the **CloudFormation deployment template** linked under **Deployment templates** (right side). Do not try to run the image directly on ECS/EKS.

3. Click the **CloudFormation deployment template** link. It opens the template in the CloudFormation console (Create stack).

Step 3 — Configure the stack

On **Specify stack details**, set the parameters:

Parameter	What to enter
Stack name	e.g. <code>refine</code>
VpcId	your VPC
SubnetId	a public subnet in that VPC
AllowedAppCidr	the CIDR allowed to reach the UI (e.g. <code>203.0.113.10/32</code>)
EnableHttps	<code>false</code> for a quick start (HTTP on port 8000); <code>true</code> if you have an ACM cert + 2 subnets
InstanceType	<code>t3.medium</code> (default) is sufficient for most deployments
AdminEmail / AdminPassword	the administrator login you want for the Refine UI (if your template version offers these — otherwise see First login below)
Everything else	leave the defaults (Watchtower auto-update on, rollback monitor on, BYOL fulfillment)

Continue through **Configure stack options** (defaults are fine), then on **Review** check "**I acknowledge that AWS CloudFormation might create IAM resources with custom names**" and choose **Submit**.

Step 4 — Wait for the stack, then read the Outputs

Wait ~8 minutes for **CREATE_COMPLETE**. Open the **Outputs** tab and note:

- `RefineUrl` — where the UI will be served (e.g. `http://<ip>:8000`).
- `DeliveryBucketName` — the S3 bucket your license goes into.
- `AdminConnectMethod` — a one-click AWS Systems Manager Session Manager link for keyless shell access to the host (no SSH key needed).

The EC2 pulls the Refine image immediately and waits for your license.

Step 5 — Activate your license (BYOL)

1. Submit the license request at <https://blacktip-ops.com/marketplace-license> with: your AWS Marketplace customer ID, the `DeliveryBucketName` from Step 4, and your contact email.

2. Blacktip Solutions emails you a signed `refine.license` file plus a one-line `aws s3 cp ...` command.
3. Run that command (from any machine with AWS access to the bucket). It copies the license into your delivery bucket.
4. An on-host watcher detects the license within ~60 seconds, installs it, and restarts the container. Refine becomes **healthy** automatically.

You can watch progress from the host via the `AdminConnectMethod` SSM link: `docker ps` (the `refine` container should go to `healthy`) and `tail -f /var/log/refine-license-watcher.log`.

Step 6 — First login

Open `RefineUrl`. Log in with your administrator credentials:

- If you set **AdminEmail** / **AdminPassword** stack parameters in Step 3, use those.
- Otherwise, your initial administrator credentials are provided by Blacktip as part of license onboarding. **Change your password immediately after first login** (Settings → Account).

Step 7 — Connect your cloud accounts

From **Accounts** → **Add Account**, connect each cloud (read-only):

- **AWS** — one-click CloudFormation onboarding stack (creates an IAM role + a Cost Explorer collector). Repeat per AWS account you want analyzed.
- **Azure** — a one-line `curl | bash` in Azure Cloud Shell (creates an Entra ID app registration, Reader role, and a Cost Management export). Auto-detects Commercial vs US Government.
- **GCP** — a Terraform module (service account + viewer roles + BigQuery billing export).

First recommendations and the cross-cloud savings ranking appear within minutes of the first sync.

Keeping Refine up to date

Auto-update is on by default: a Watchtower sidecar polls AWS Public ECR on a schedule (default daily 03:00 UTC) and updates the Refine container in place when a new version is published — with an automatic rollback to the last-known-good image if a new build fails its health check. To pin a version, set `EnableAutoUpdate=false` and a specific `RefineImageUri` tag.

Troubleshooting

Symptom	Check
UI won't load	Confirm your current IP matches AllowedAppCidr ; for HTTP, the URL is <code>http://<ip>:8000</code> .
"License" / waiting screen	The license hasn't landed yet — confirm the <code>aws s3 cp</code> ran against the correct DeliveryBucketName ; watch <code>/var/log/refine-license-watcher.log</code> via SSM.
Container keeps restarting	Expected before the license is installed; it stops once a valid <code>refine.license</code> is in the bucket.
Need a shell on the host	Use the AdminConnectMethod SSM Session Manager link from the stack Outputs — no SSH key required.

Support

- **Setup / licensing:** support@blacktip-ops.com
- **Sales / tiers:** sales@blacktip-ops.com